

野火_H750X_H743X_底板_原理图_V1.0

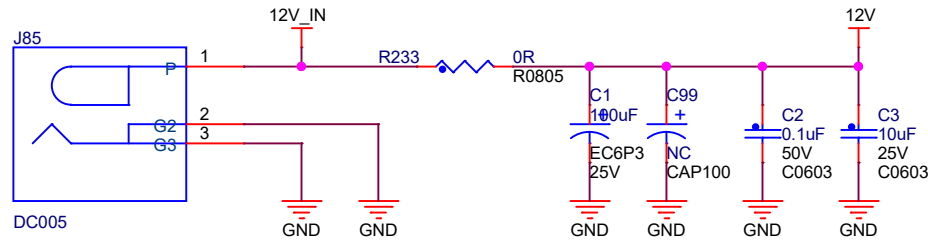
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历史版本

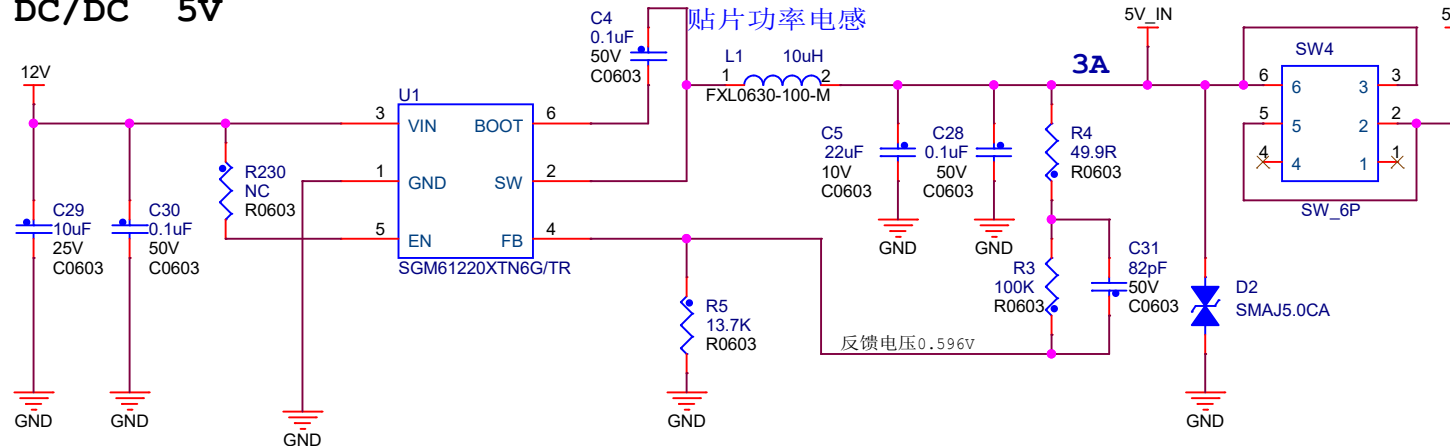
版本号	日期	设计	描述
V1.0	2018-12-06	cancore	初始版本
V1.1	2021-09-27	cancore	修改电源芯片，USB转串口换成CH340C，修改USB接口HOST，Device各一个

DC IN : 6V~12V



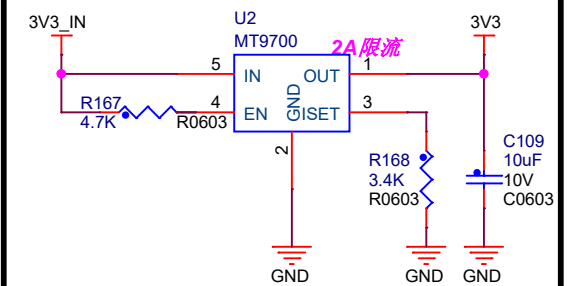
外部电源供电时断开USB供电，
防止灌电流到电脑USB，保护电脑USB口

DC/DC 5V

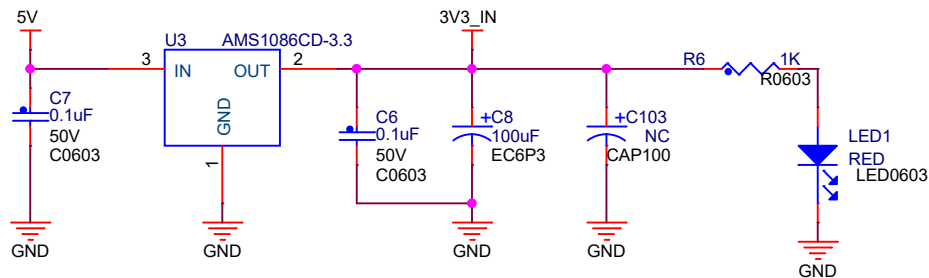


R230为100K/NC
根据使用芯片决定是否需要焊对应的电阻

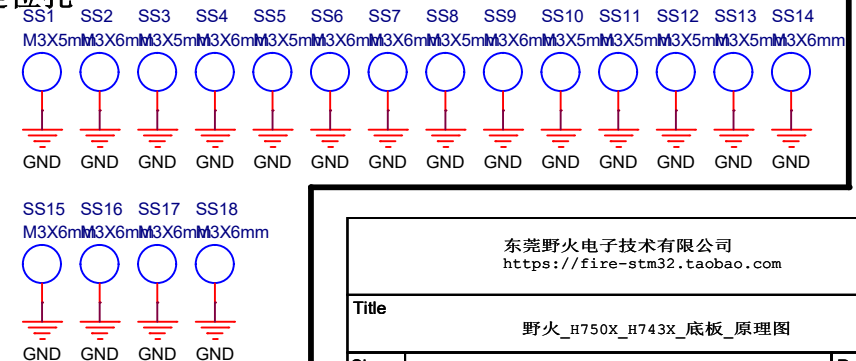
3.3v 输出过流保护



5v转3.3v



定位孔



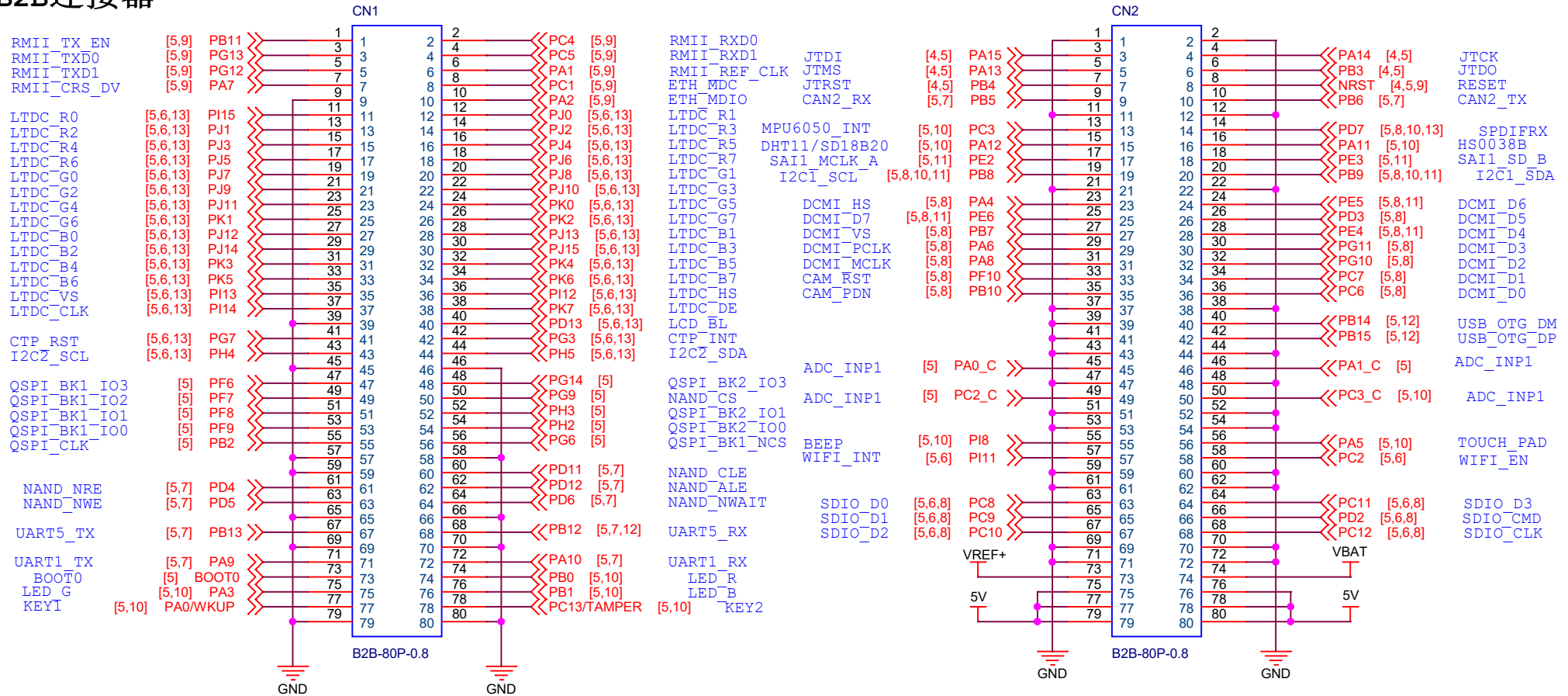
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Title
野火_H750X_H743X_底板_原理图

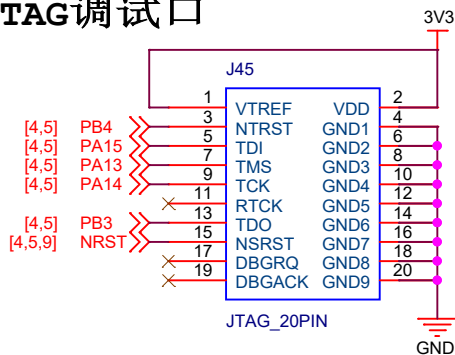
Size A4 Document Number 主电源 Rev V1.0

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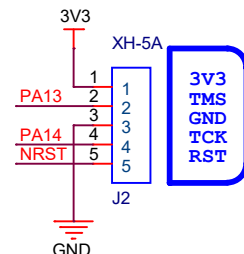
B2B连接器



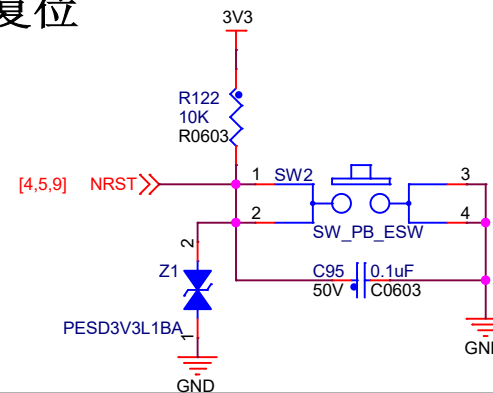
JTAG调试口



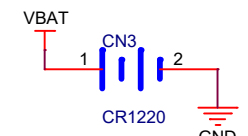
SWD调试口



复位



RTC电池



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Title

野火_H750X_H743X_底板_原理图

Size
A4

Document Number
MCU_B2B

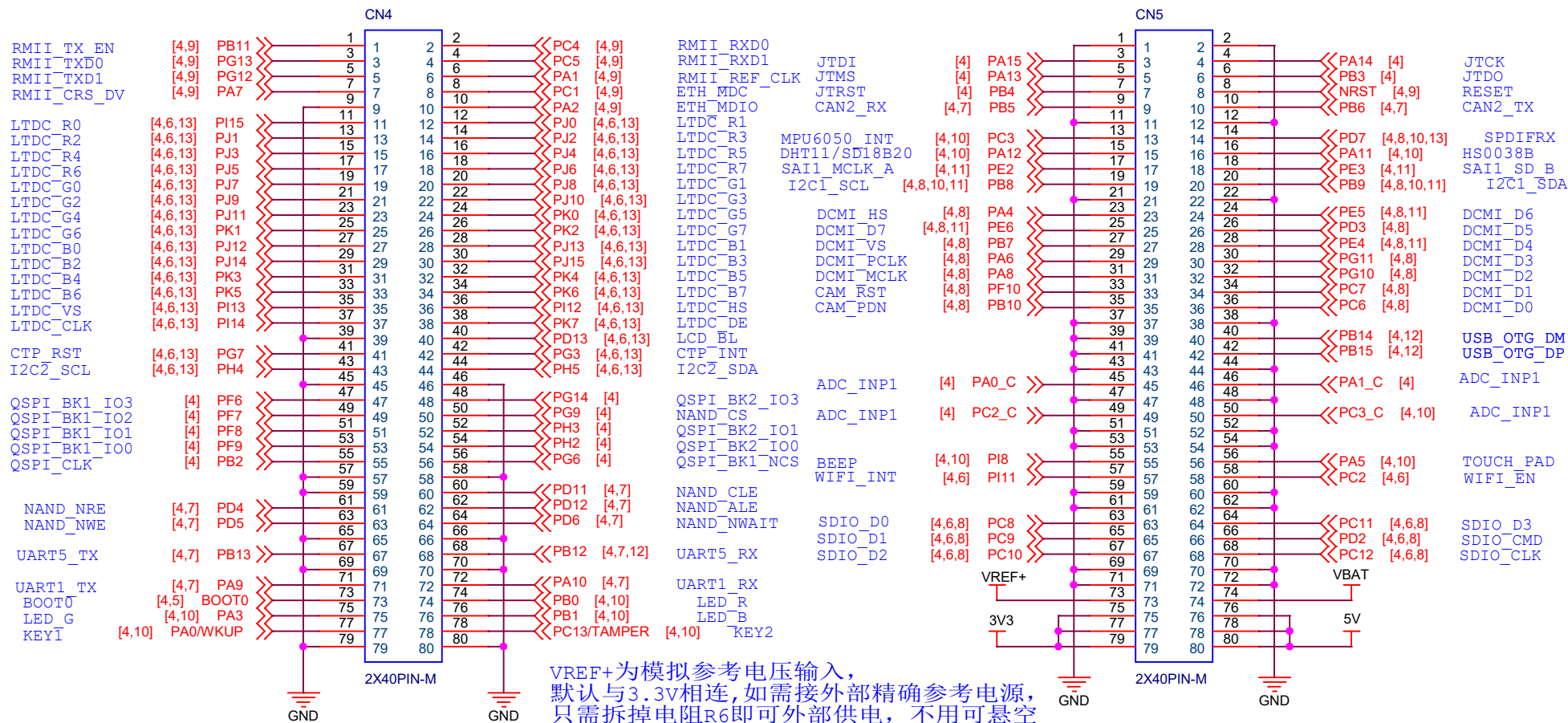
Rev
V1.0

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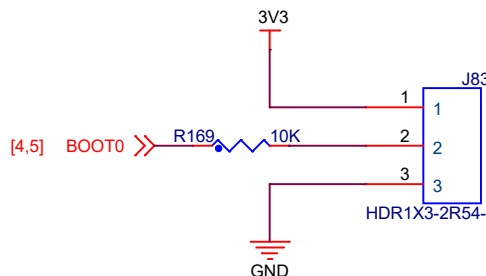
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引出IO口(排针)

GPIO引出107个, 模拟专用口4个, SDRAM部分除外, 共57个未引出。
如果将SDRAM的IO引出, SDRAM必须降频使用, 发挥不了最佳性能, 这样得不偿失。



设置启动模式



H750X/H743X BOOT设置

BOOT0	BOOT1	启动方式
0	BOOT_ADD0[15:0]	启动地址由用户选择字节 BOOT_ADD0[15:0] ST 编程值定义: Flash 位于 0x0800 0000
1	BOOT_ADD1[15:0]	启动地址由用户选择字节 BOOT_ADD1[15:0] ST 编程值定义: 系统自举程序位于 0x1FF0 0000

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Title

野火_H750X_H743X_底板_原理图

Size

Document Number

引出IO/BOOT

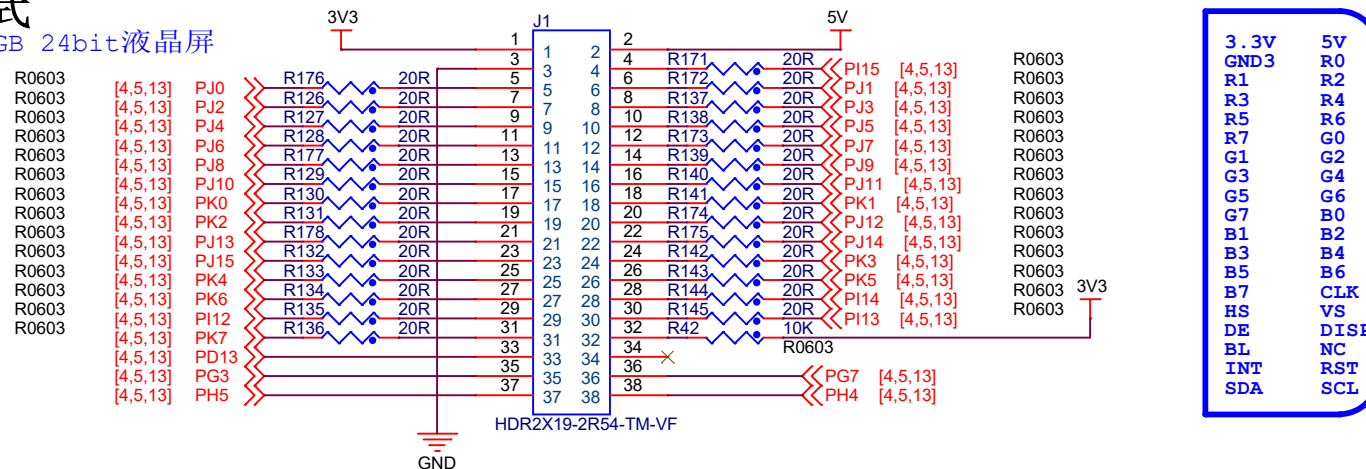
Rev

V1.0

Date: Wednesday, October 23, 2024

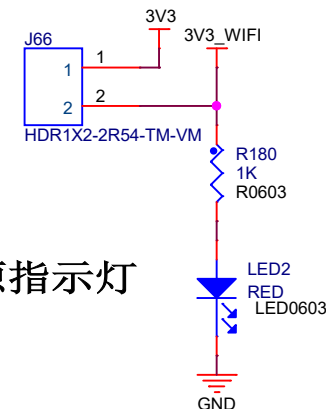
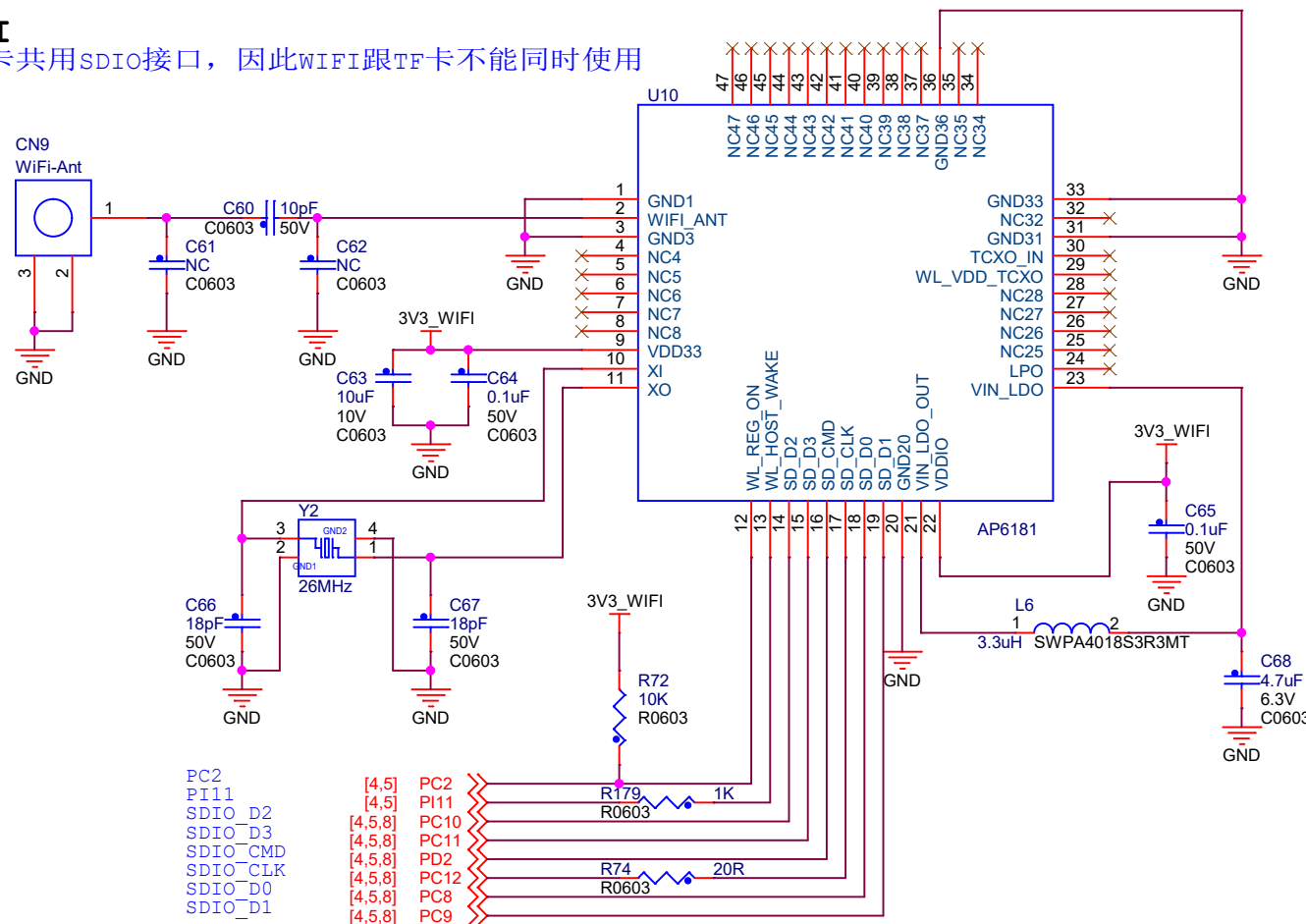
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支持高达1024*768分辨率的RGB 24bit液晶屏



WIFI

与TF卡共用SDIO接口，因此WIFI跟TF卡不能同时使用



WIFI电源指示灯

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Title	野火 H750X H743X 底板 原理图
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Size
A4

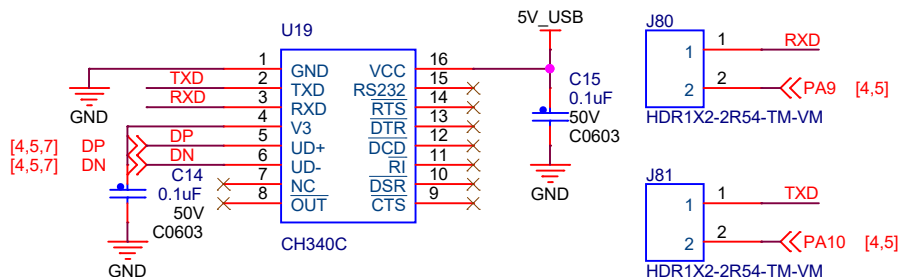
Document Number	液晶屏/WIFI
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Rev	V1.0
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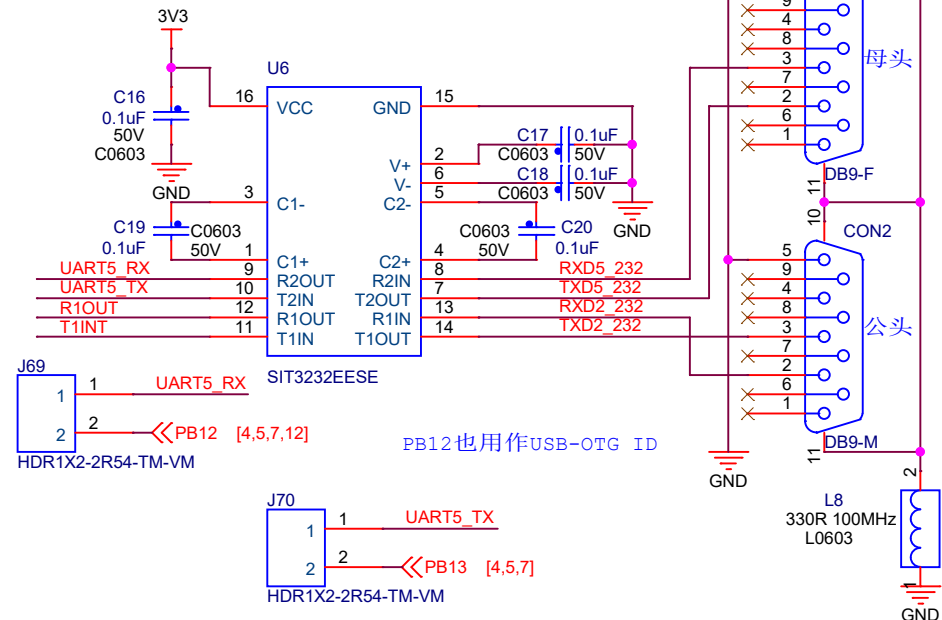
Date: Friday, October 25, 2024

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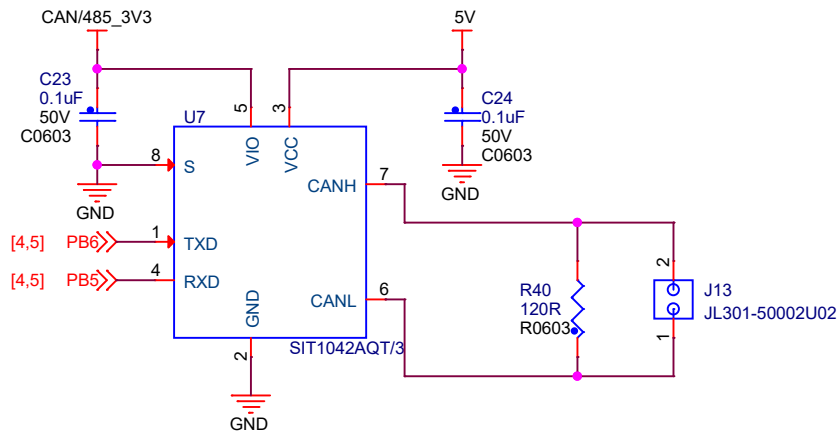
USB转串口



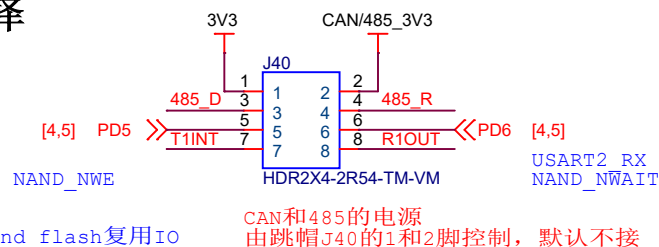
RS232串口



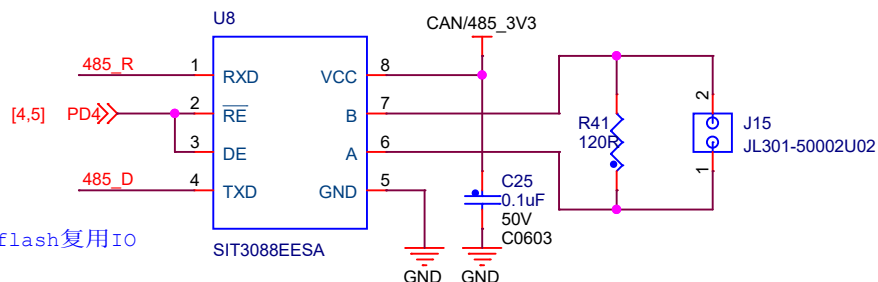
CAN



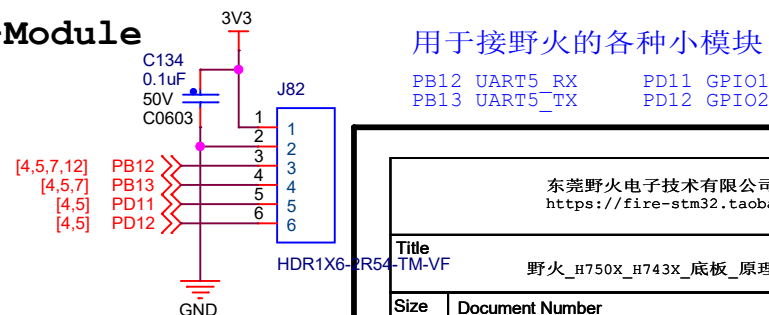
跳帽选择



RS485



EBF-Module



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野火_H750X_H743X_底板_原理图

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摄像头接口

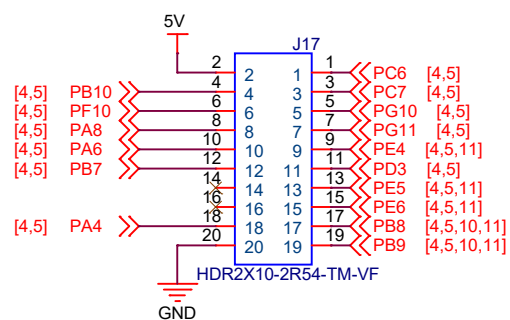
像素时钟最大支持80MHz

内嵌码/外部行同步和帧同步

支持连续模式或快照模式，裁剪功能

支持以下数据格式：

- 1、8 位逐行视频：单色或原始拜尔格式
- 2、YCbCr 4:2:2 逐行视频
- 3、RGB 565 逐行视频
- 4、压缩数据： JPEG

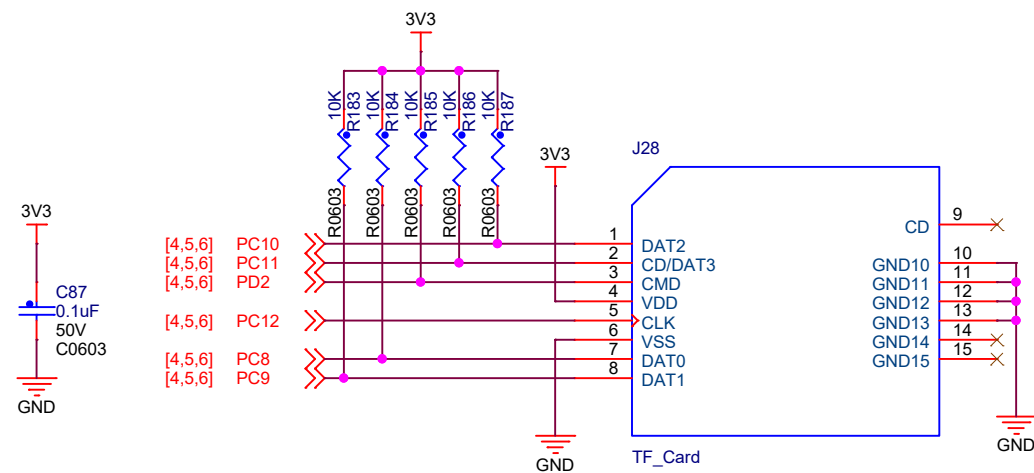


VDD	D0
PDN	D1
RST	D2
CLKIN	D3
PCLK	D4
VSNC	D5
NC13	D6
NC15	D7
HSYNC	SCL
GND	SDA

默认PA8悬空，摄像头时钟由有源晶振提供

SD卡

与WIFI复用SDIO引脚



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Document Number

Camera/SD/SPDIF

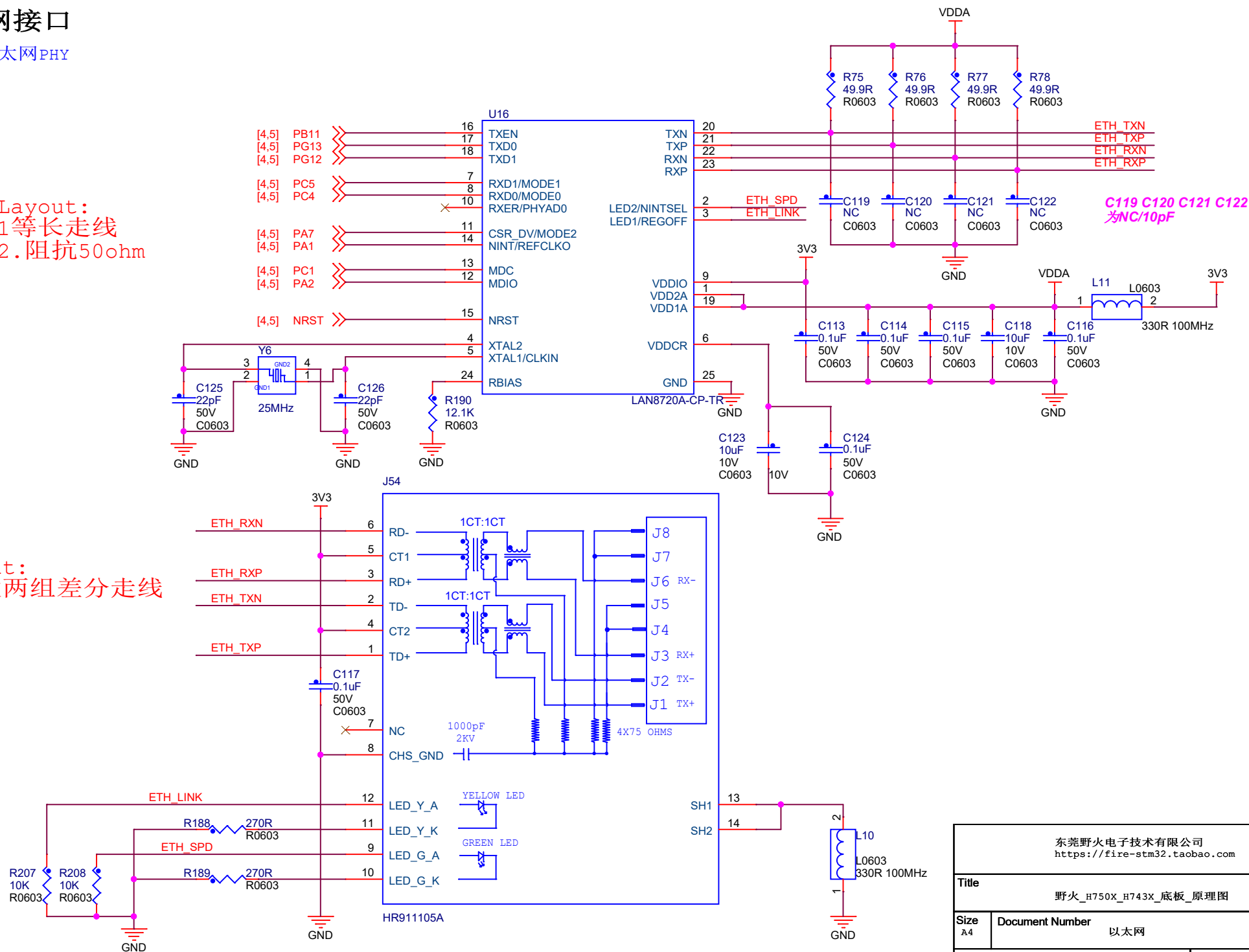
Rev
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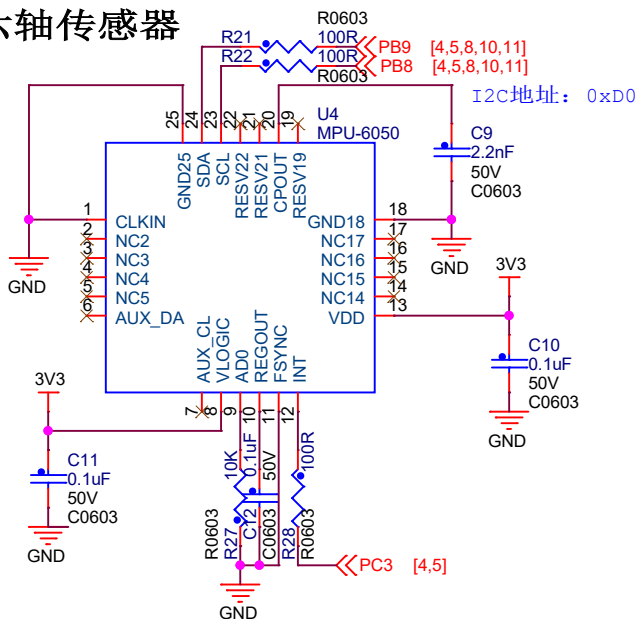
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百兆以太网PHY

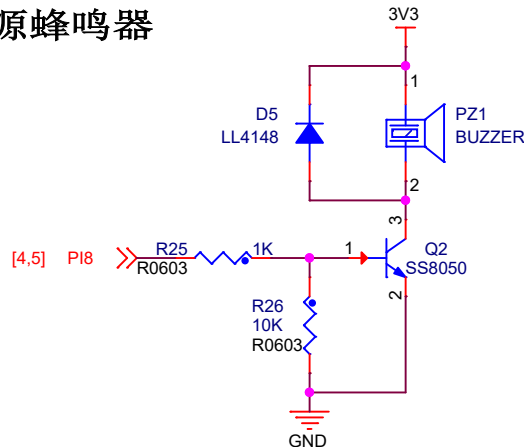
Layout:
1收发两组差分走线



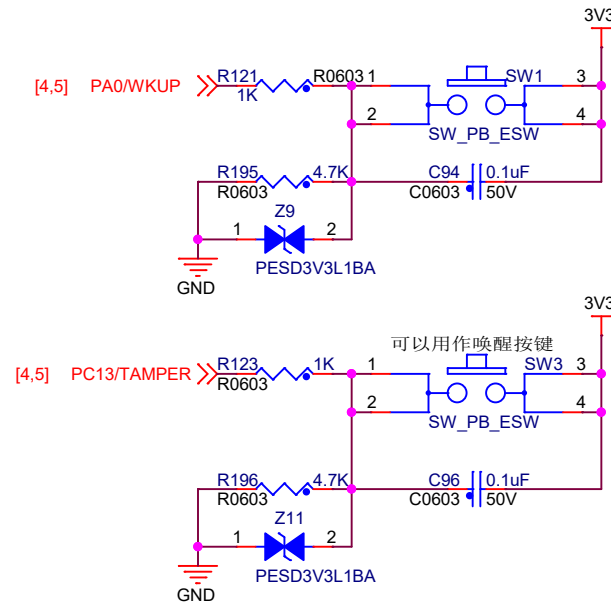
六轴传感器



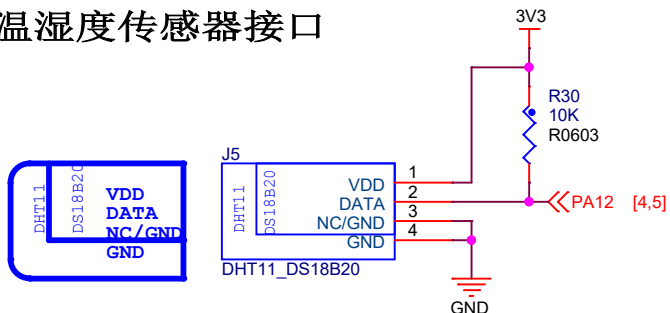
有源蜂鸣器



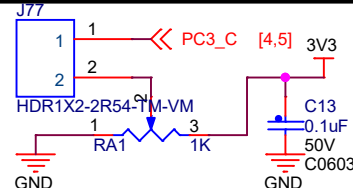
按键



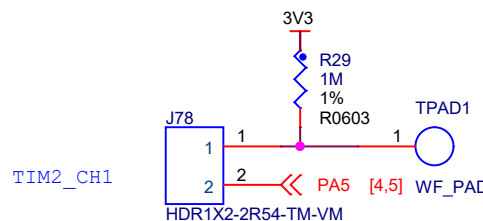
温湿度传感器接口



可调电阻

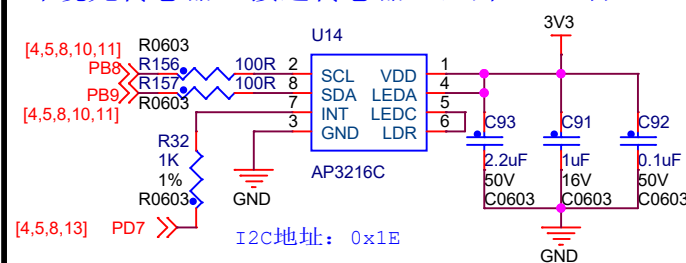


触摸按键

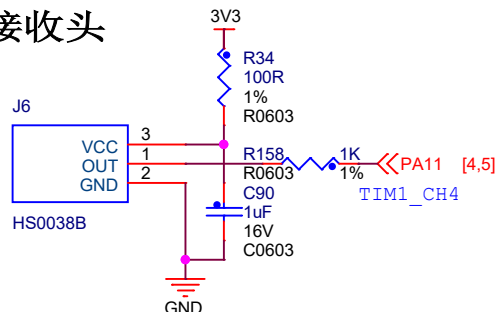


ALS+PS+IRLED (三合一)

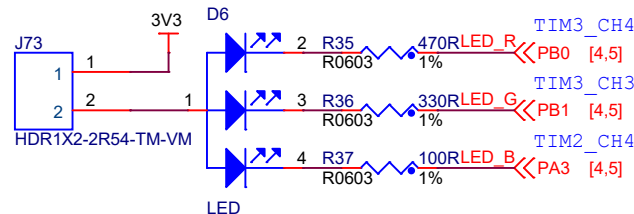
环境光传感器、接近传感器、红外LED三合一



红外接收头



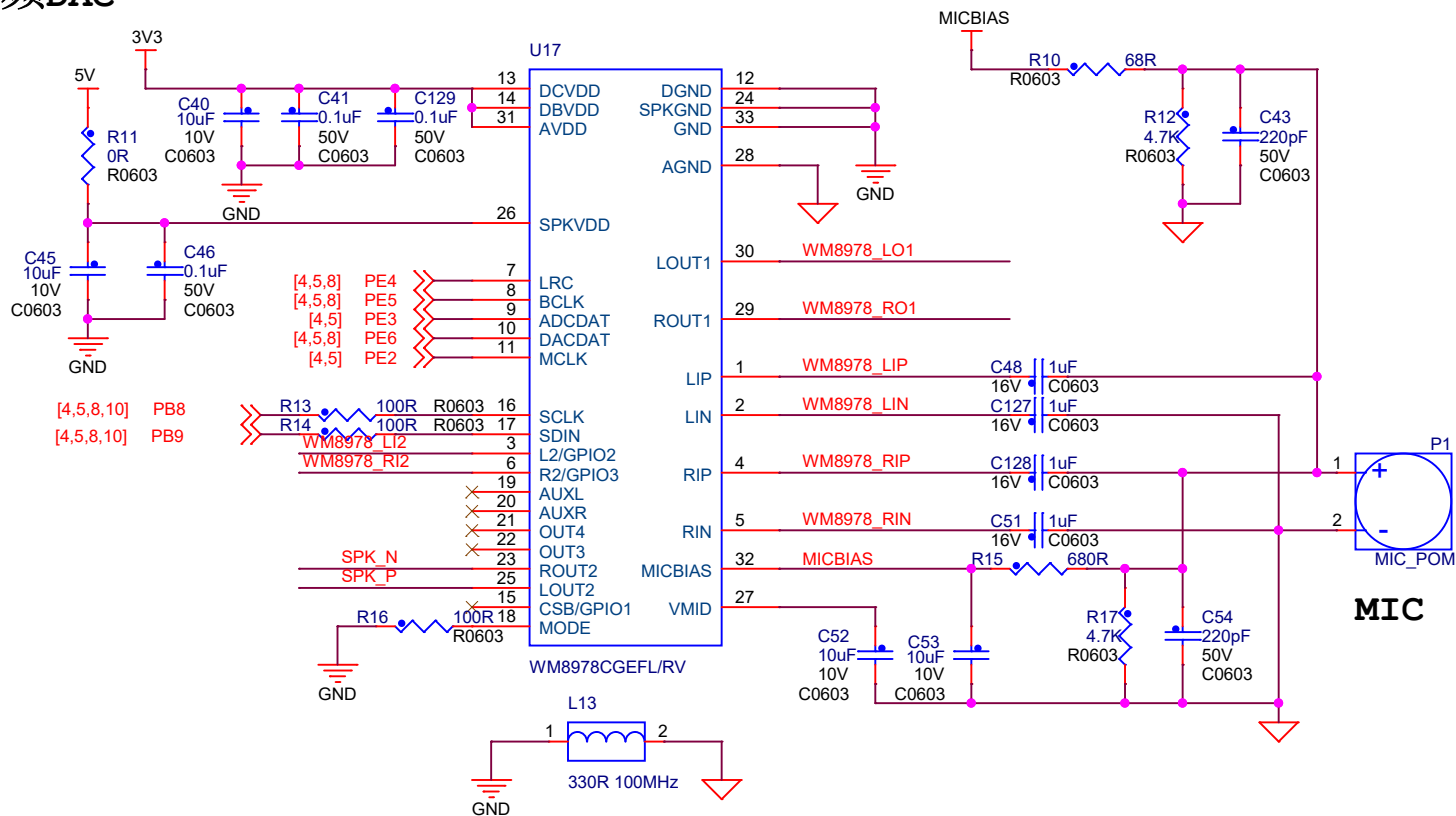
彩色LED



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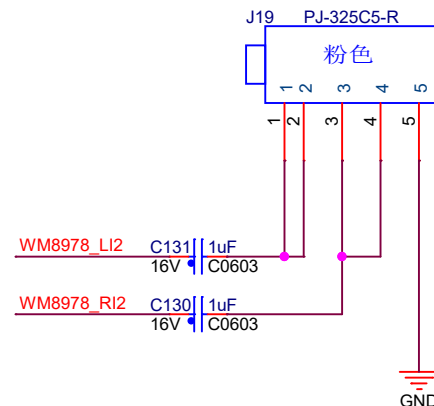
Title		
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SENSOR/KEY/LED		
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音频DAC

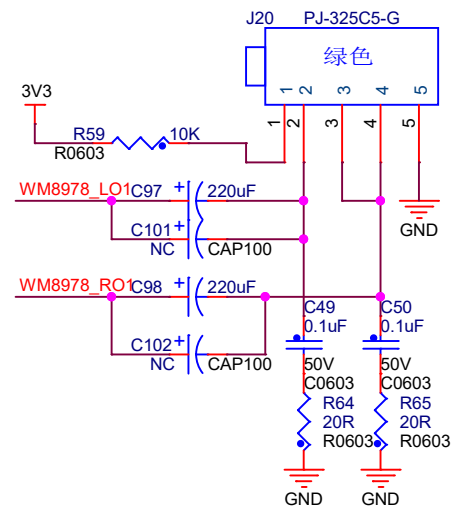


SAI1与DCMI接口复用IO
所以摄像头与音频不能同时使用

音频输入

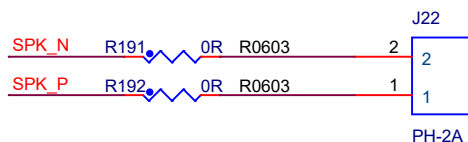


耳机插座



喇叭插座

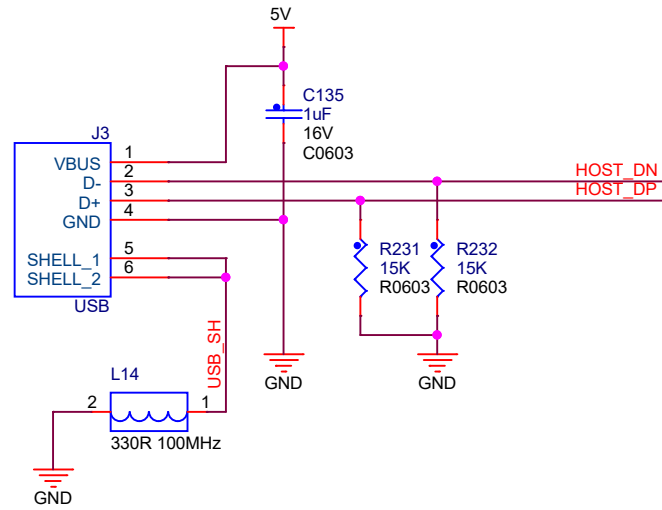
可以驱动8Ω 1W的喇叭（BTL方式）



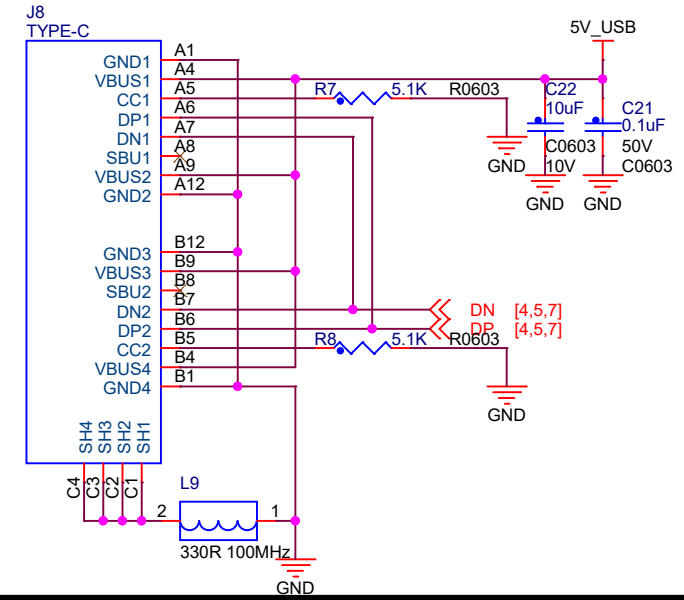
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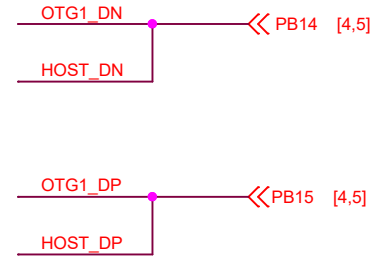
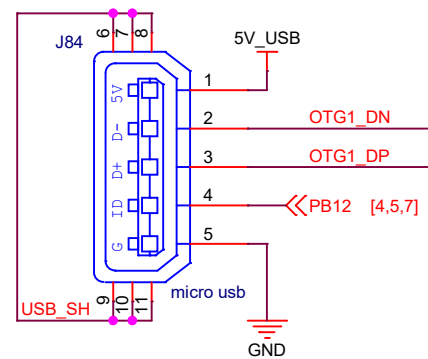
USB HOST



USB TO UART



USB Device



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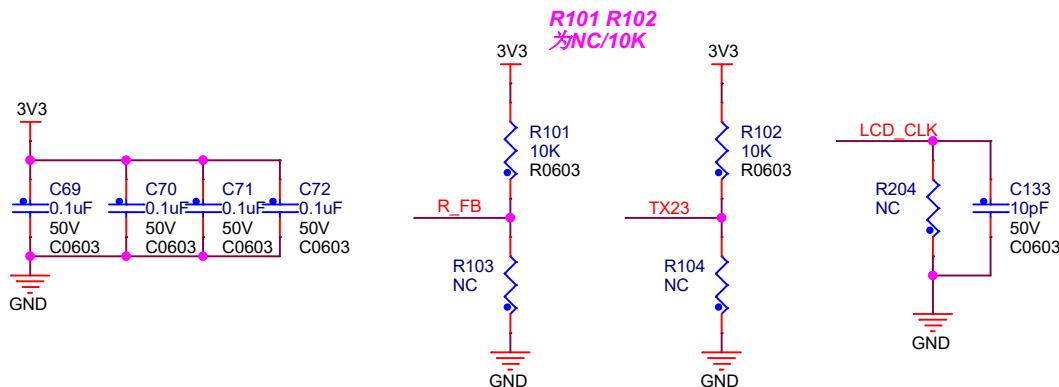
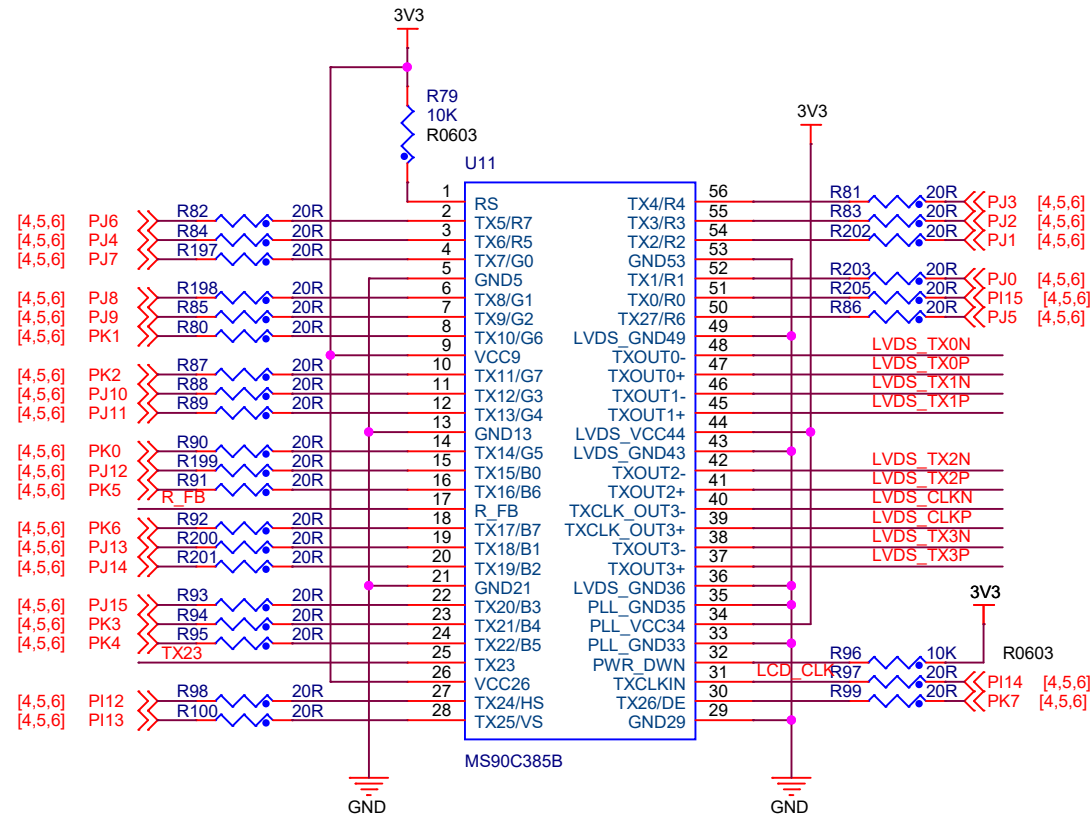
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A4	USB	V1.0

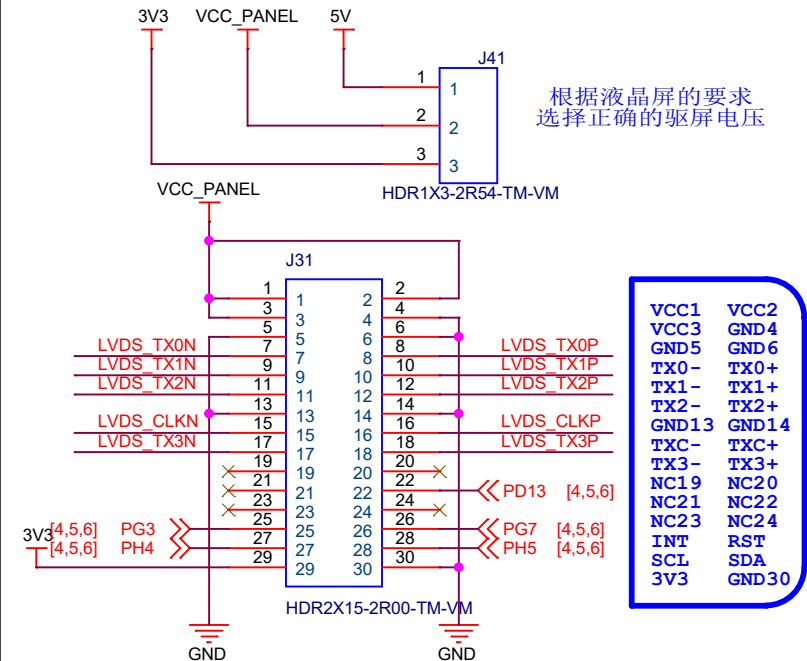
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RGB888转LVDS

支持高达1024*768分辨率
支持单八位的LVDS接口的液晶屏



驱屏接口



液晶背光接口

